

Qns	Answer	Marks
3(a)(i)	$E = mL_v + h_{\text{loss}} = Pt$ $P = \left(\frac{m}{t}\right)L_v + \frac{h_{\text{loss}}}{t}$ $\left. \begin{aligned} P_1 &= \left(\frac{m}{t}\right)_1 L_v + \left(\frac{h_{\text{loss}}}{t}\right)_1 \quad \text{--- (1)} \\ P_2 &= \left(\frac{m}{t}\right)_2 L_v + \left(\frac{h_{\text{loss}}}{t}\right)_2 \quad \text{--- (2)} \end{aligned} \right\}$ $(1) - (2)$ $P_1 - P_2 = \left[\left(\frac{m}{t}\right)_1 - \left(\frac{m}{t}\right)_2 \right] L_v$ $L_v = \frac{P_1 - P_2}{\left(\frac{m}{t}\right)_1 - \left(\frac{m}{t}\right)_2} = \frac{I_1 V_1 - I_2 V_2}{\left(\frac{m}{t}\right)_1 - \left(\frac{m}{t}\right)_2}$ $= \frac{(5.0)(78) - (4.0)(60)}{\frac{16 \times 10^{-3}}{1.5 \times 60} - \frac{10 \times 10^{-3}}{1.5 \times 60}}$ $= 2.25 \times 10^6 \text{ J kg}^{-1}$	1 1 A0
3(a)(ii)	pure substances undergo phase change at constant temperature (words to that effect) so temperature difference with surrounding kept constant	1
3(b)(i)	$Q = mL$ $= (1.0)(2.25 \times 10^6)$ $= 2.25 \times 10^6 \text{ J}$	1 1
3(b)(ii)	$w_{\text{on}} = -p\Delta V = -p(V_{\text{final}} - V_{\text{initial}})$ $= -(1.0 \times 10^5)(1.67 - 1.04 \times 10^{-3})$ $= -(1.0 \times 10^5)(1.66896)$ $w_{\text{by}} = +1.67 \times 10^5 \text{ J}$	1 1
3(b)(iii)	$\Delta U = Q + W$ $= 2.25 \times 10^6 + (-1.67 \times 10^5)$ $= 2.08 \times 10^6 \text{ J}$	1
3(b)(iv)	$\Delta PE = PE_{\text{gas}} - PE_{\text{liquid}}$ $PE_{\text{gas}} = \Delta PE + PE_{\text{liquid}}$ $= 2.08 \times 10^6 + 3.41 \times 10^5$ $= 2.42 \times 10^6 \text{ J}$	1 1

Qns	Answer	Marks
3(b)(v)	pure substances undergo phase change at constant temperature so total translational kinetic energy of particles remain constant	B0
	large increase in volume for phase change from liquid to gas so separation between molecules instead	1
	intermolecular bonds are complete broken as potential energy between particles increases	1
	work is done against atmosphere	1

Qns	Answer	Marks
4(a)(i)	current in solenoid produces magnetic field with field lines parallel to axis	1